

Luca Daloia

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Education:

North Carolina State University, Raleigh, NC – Dec 2026

BS - Electrical and Computer Engineering (Double Major) - Cumulative GPA: 3.39/4.0

Work Experience:

Summer 2025 - NC State Univ (Research Support for Prof. John Muth, PhD - Progress Energy Distinguished Prof)

Developed a Django-based web application to organize, visualize, and compare semiconductor device data from research publications. The platform supports structured device records, advanced filtering, and interactive performance benchmarking.

- Implemented interactive Baliga's Figure of Merit (BFOM) plots that allow Prof. Muth to instantly compare internal research results against material limits and competitors worldwide.
- Enabled a web based hyperlink on each point in the BFOM plot to a detailed device page containing parameters, original research plots, structure images, and references to review and source papers.

Summer 2024 – Corning Inc. (Division Controls Engineer Intern)

Developed a suite of Python tools to enhance extruder performance monitoring and data management.

- Automated data extraction from servers into Excel, reducing manual effort.
- Built analysis scripts to evaluate extruder heating performance and identify inefficiencies.
- Created a GUI (using Tkinter) tool with options for data import, filtering, comparison, and visualization.
- Authored documentation including installation instructions, user guide, and code references.

Summer 2023 – Daloia Consulting (Personal Project)

Developed a Python program with Tkinter (GUI) and SQLite (database) to calculate the smallest conduit size meeting NEC (National Electrical Code) requirements. The tool stored user inputs in an editable database and streamlined calculations, saving time and reducing errors.

Relevant Coursework:

Senior Design - Auger Alignment System (Sponsored by Caterpillar Inc.)

Engineered a comprehensive wireless alignment system for hydraulic augers by designing a custom multi-layer PCB in KiCad that integrated ESP32 microcontrollers with high-precision IMU sensors. Architected the full hardware solution from schematic capture to final layout, incorporating dedicated power management stages and optimized sensor interfaces for industrial reliability. Developed the underlying firmware in C to process raw inertial data and deliver real-time orientation feedback via Bluetooth, utilizing an intuitive LED interface to ensure precise vertical alignment for operators in the field.

ECE 306 – Introduction to Embedded Systems

Designed and implemented a TI MSP430-based autonomous robotic platform featuring integrated power regulation, H-bridge motor control, and multi-sensor fusion for precision navigation. Programmed the embedded system in C to process real-time ADC data from IR emitter-detector arrays, enabling the vehicle to detect and track physical course boundaries and transition between autonomous line-following and manual operation. Developed a full-stack IoT solution by integrating a Wi-Fi module and a web-based steering interface, while verifying all hardware subsystems using Analog Discovery and digital voltmeters to ensure signal integrity and power stability.

ECE 308 – Introduction to Control Systems

Designed and simulated a self-balancing Segway control system in MATLAB. Derived system equations from DC motor and inverted pendulum equations, modeled transfer functions and state-space representation, and implemented a full-state feedback controller for stabilization.

ECE 492 - Embedded Systems Hardware Design

Engineered a microcontroller-based embedded system by interpreting manufacturer schematics to design a custom power circuit validated through Spice simulations. Using KiCad, I developed a multi-layer PCB integrating LDRs, LEDs, and tactile buttons while managing complex routing, power planes, and mechanical constraints.

Other Skills:

- Technical Skills: Python (Django, Tkinter), C++, C, MATLAB, LaTeX, Embedded Systems, Control Systems, Schematic and PCB Design, ESP32, MSP430, Excel, ChatGPT, Google Gemini, Microsoft Copilot, Github
- Invited to apply to the NC State Honors Program
- Dean's List (2022, 2023, 2024)
- Soft skills: emotional intelligence, leadership, responsibility, creativity, team spirit, problem-solving